

Q.No:5

Problem Description

You are given a string which comprises of lower case alphabets (a-z), upper case alphabets (A-Z), numbers, (0-9) and special characters like !,.,; etc.

You are supposed to find out **which character occurs the maximum number of times and the number of its occurrence**, in the given string. If two characters occur equal number of times, you have to output the character with the lower **ASCII value**.

For example, if your string was: **aaaaAAAA**, your output would be: **A 4**, because **A** has lower ASCII value than **a**.

Input format:

The input will contain a string.

Output format:

You've to output two things which will be separated by a space:

- i) The character which occurs the maximum number of times.
- ii) The number of its occurrence.

Constraints:

The maximum length of the string can be 1000.

Logical Test Cases

Test Case 1

INPUT (STDIN)

puppy is a dog!!!!!!!!!!!!

EXPECTED OUTPUT

! 12

Test Case 2

INPUT (STDIN)

leana is a singer!!!!!!!!

EXPECTED OUTPUT

! 7

Mandatory Test Cases

Test Case 1

KEYWORD

for(i=0;i<l;i++)

Code:

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[1001];
    int freq[256] = {0};
    fgets(str, sizeof(str), stdin);
    int len = strlen(str);
    if (len > 0 && str[len - 1] == '\n') {
        str[len - 1] = '\0';
    }
    for (int i = 0; str[i] != '\0'; i++) {
        freq[(unsigned char)str[i]]++;
    }
}
```

```
}  
int max = 0;  
char ch;  
for (int i = 0; i < 256; i++) {  
    if (freq[i] > max) {  
        max = freq[i];  
        ch = i;  
    }  
}  
printf("%c %d\n", ch, max);  
  
return 0;  
}
```

Output:

Test_case:1

```
cisc7/graph/55  
puppy is a dog!!!!!!!!!!!!  
Output:  
! 12  
aaditvaiaiswal@Aadityas-MacBook
```

Test_case:2

```
leana is singer!!!!!!  
Output:  
! 7  
aaditvaiaiswal@Aadityas-MacBook
```